Customer Shopping Behavior

Analysis & Dashboard Report — 3,900 Customer Records | 19 Features

PYTHON · POSTGRESQL · POWER BI

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Executive Summary

$233K

Total Revenue

$59.76

Avg Purchase

3.75

Avg Rating (out of 5)

3,900

Customer Records

Gender Split

Male customers: **68%** of base, **67.7%** of revenue

Top Category

Clothing: **$104,264** — 44.7% of total revenue

Subscriptions

Only **27%** of customers hold an active subscription

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Project Overview

Objective

Analyze customer shopping behavior of a shop to uncover patterns in demographics, product preferences, purchase frequency, discount usage, and revenue — presented through an interactive Power BI dashboard.

Dataset

3,900 records across 19 columns: demographics, purchase details, behavioral attributes, and satisfaction metrics including age group, category, payment method, and review rating.

Python / Pandas

Cleaning, preprocessing, feature engineering

PostgreSQL + SQLAlchemy

Structured querying via Colab

Power BI Desktop

Interactive dashboard & KPI cards

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Data Cleaning & Preprocessing

01

Missing Values

Review ratings filled using column mean, then refined with category-level median via `groupby()` and `transform()`.

02

Feature Engineering

Created `age_group` (4 quantile bins) and `purchase_frequency_days` (numeric conversion of purchase cadence).

03

Redundancy Removal

`promo_code_used` was confirmed identical to `discount_applied` and dropped to eliminate duplication.

04

Data Type Verification

Numeric columns confirmed as correct types; categorical columns stored as text for PostgreSQL and Power BI compatibility.

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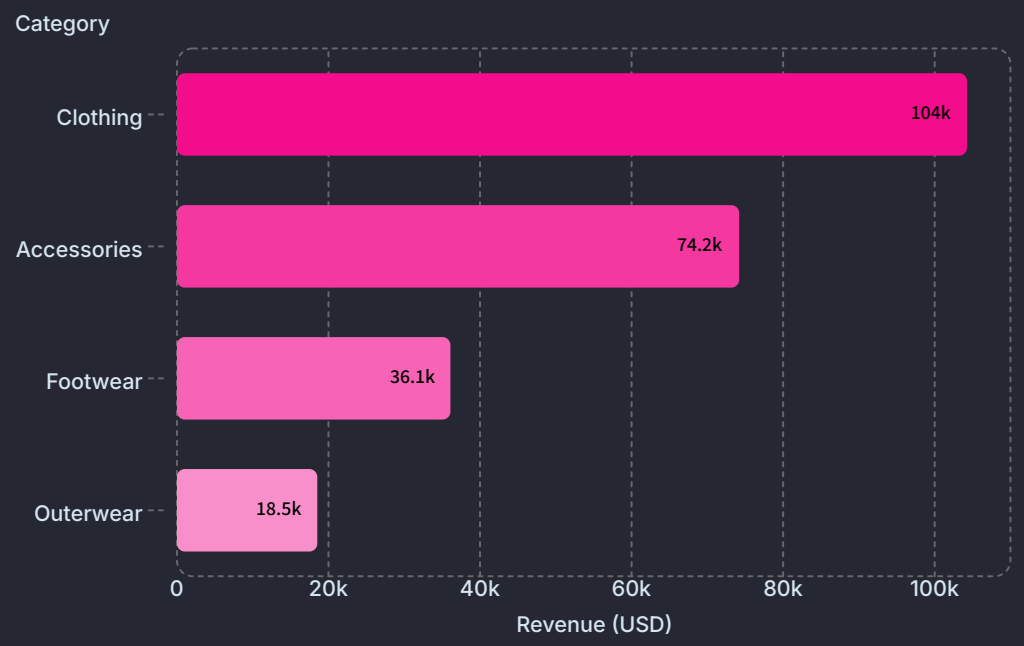
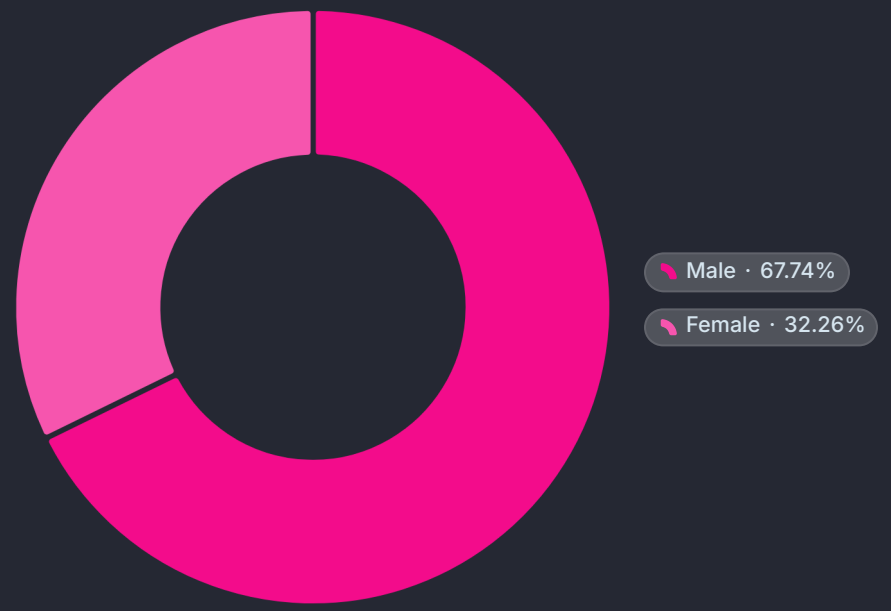
Revenue by Category & Gender



Gender Revenue Split



Male customers (2,652) generated **$157,890** vs. female customers (1,248) at **$75,191**.

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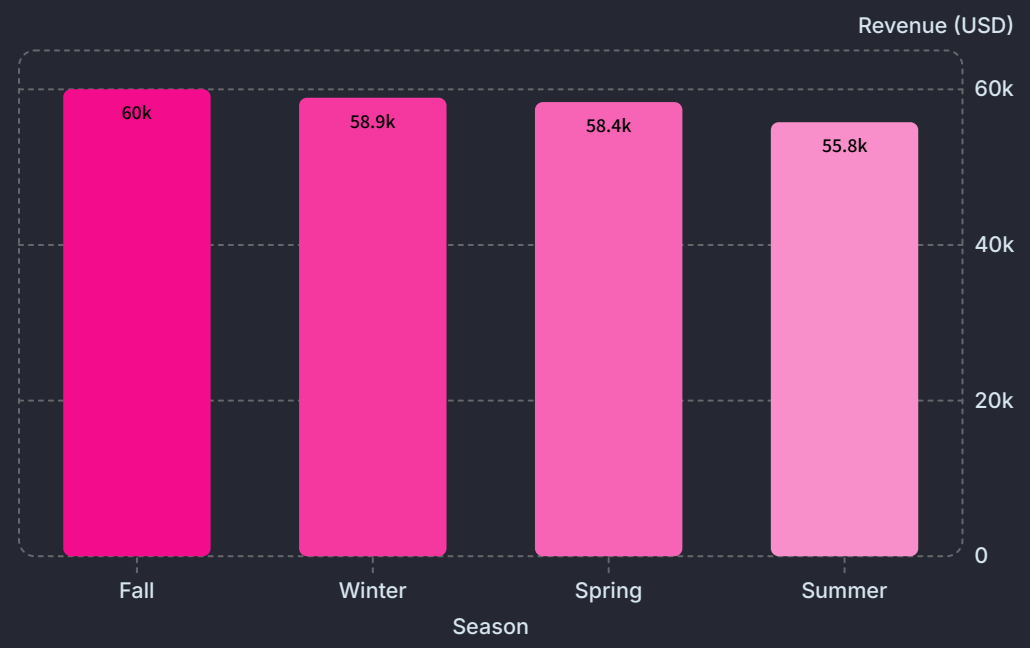
Seasonal Revenue & Top Items



Top 5 Items by Revenue

Item	Revenue
Blouse	$10,410
Shirt	$10,332
Dress	$10,320
Pants	$10,090
Jewelry	$10,010

Top items are tightly clustered in the **$10,000–$10,500** range, indicating balanced product performance.

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Key Findings & Insights

Demographics

Age groups are evenly distributed. Only **27%** hold subscriptions — a major growth opportunity.

Category Dominance

Clothing + Accessories = **77%** of revenue. Clothing alone is 1.4× the next category.

Price Sensitivity

43% of purchases used a discount, signaling high price sensitivity across the customer base.

Satisfaction

Avg rating **3.75/5** across all categories (3.72–3.79), indicating consistent customer satisfaction.

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Power BI Dashboard

KPI Cards

Total Revenue · Avg Purchase Amount · Avg Review Rating

Visualizations

- Revenue by Gender (Bar), Category (Pie/Bar), Season (Column)
- Avg Rating by Item · Purchase by Age Group · Discount vs Revenue

Interactive Slicers

Filter all visuals by **Gender, Category, Season, Discount Applied,** and **Age Group** simultaneously.

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Recommendations

Grow Subscriptions

73% are non-subscribers. Target them with exclusive discounts and loyalty rewards.

Double Down on Clothing & Accessories

These two categories drive 77% of revenue — expand lines and offer seasonal promotions.

Close the Summer Gap

Fall leads revenue; Summer lags. Targeted campaigns and new summer launches can reduce the gap.

Engage Female Customers

Female customers are only 32% of the base. Tailored marketing in top categories can grow this segment.

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Technology Stack & Conclusion

Python / Pandas

Cleaning, feature engineering

Google Colab

Cloud notebook environment

PostgreSQL + SQLAlchemy

Structured querying

Power BI Desktop

Interactive dashboard

Final Takeaway

Analysis of **3,900 records** reveals a male-dominated, price-sensitive customer base with strong preference for Clothing and Accessories. With only **27% subscribed**, the largest opportunity lies in retention, subscription growth, and targeted female engagement.

End-to-end pipeline: Python → PostgreSQL → Power BI

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